
Local MAP Sampling for Diffusion Models

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Abstract

Diffusion Posterior Sampling (DPS) provides a principled Bayesian approach to inverse problems by sampling from $p(x_0 | y)$. While posterior sampling is valuable for capturing uncertainty and multi-modality, many classical and practical inverse problem settings ultimately prioritize accurate point estimation—most notably the MAP estimator, which has long served as a standard reconstruction objective in imaging and scientific applications. We introduce *Local MAP Sampling (LMAPS)*, a new inference framework that iteratively solving local MAP subproblems along the diffusion trajectory. This perspective clarifies their connection to global MAP and DPS, offering a unified probabilistic interpretation for optimization-based methods. Building on this foundation, we develop practical algorithms with a covariance approximation motivated by Gaussian prior assumption, a reformulated objective for stability and interpretability. Across a broad set of image restoration and scientific tasks, LMAPS achieves state-of-the-art performance.

1. Introduction

Diffusion Posterior Sampling (DPS) is a recently proposed framework that extends diffusion generative models to Bayesian inference (Chung et al., 2022; Song et al., 2023c). This framework is particularly powerful for a wide range of applications, ranging from combined guidance and style transfer (Ye et al., 2024) to inverse problems such as medical imaging (Chung & Ye, 2022), image restoration (Chung et al., 2022), and scientific data reconstruction (Zheng et al., 2025), where it enables high-quality reconstructions while also providing principled uncertainty quantification (Ye et al., 2024). DPS conditions the generative process on observed measurements, enabling efficient sampling from

posterior distributions over clean data $p(x_0 | y)$. This group of approaches and variants includes but not limited to TMPD (Boys et al., 2023), DDNM (Wang et al., 2022), IIGDM (Song et al., 2023b), TFG (Guo et al., 2025).

While posterior sampling is fundamentally important in Bayesian inverse problems—capturing multi-modality, providing calibrated uncertainty, and supporting downstream decision making through credible intervals and risk-sensitive criteria—there is a parallel and long-standing line of work that emphasizes point estimation, and in particular MAP, as an equally central objective. Classical treatments of Bayesian inverse problems show that the MAP estimator often coincides with the solution of a variationally regularized optimization problem and is widely used as a practical reconstruction rule in imaging, medical, and geophysical applications (Stuart, 2010; Kaipio & Somersalo, 2005; Tarantola, 2005).

Optimization-based approaches—such as Resample (Song et al., 2023a), DiffPIR (Zhu et al., 2023), DCDP (Li et al., 2024), and DMPlug (Wang et al., 2024)—have shown strong performance by alternating between denoising, optimization, and resampling to address inverse problems. Unlike DPS, which attempts to sample from the posterior distribution $p(x_0 | y)$, optimization-based approaches prioritize reconstruction performance over distributional faithfulness. Nevertheless, it’s still unclear if the iterative procedure converges to the global MAP solution, i.e., $\arg \max p(x_0 | y)$, would it still be consistent with DPS? Clarifying this foundation could provide both a principled interpretation and a stronger theoretical basis for optimization-based methods.

In this work, we argue that the optimization steps in these methods inherently solve a *local MAP problem*. But the resulting solutions neither converge to the global MAP nor equivalent to posterior sampling. Instead, they are more likely to reflect a trade-off between the two.

Our main contributions are summarized as follows:

- **Theoretical.** We formulate *Local MAP Sampling (LMAPS)*, a new inference framework that iteratively solves local maximum-a-posteriori subproblems along the diffusion trajectory. LMAPS is closely related to DMAP (Xu et al., 2025); our clean-signal-space formulation provides a complementary view that makes the

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local objective and its optimization structure explicit. We analyze its relationship to global MAP and DPS, and show that LMAPS unifies Tweedie Moment Projected Diffusion (TMPD) and optimization-based inverse problem methods under a single framework. The relationship between LMAPS and existing methods are presented in Figure 1.

- **Methodological.** To address inverse problems, we introduce a covariance approximation motivated by Gaussian prior assumption. In addition, we propose an objective reformulation that improves interpretability and enhances numerical stability.
- **Empirical.** LMAPS is validated on 10 image restoration tasks (linear, nonlinear, non-differentiable) and 3 scientific inverse problems. It achieves the best results in 43/60 FFHQ/ImageNet PSNR/SSIM/LPIPS cases, while being more efficient than DAPS. On scientific tasks, LMAPS consistently attains the highest PSNR, including > 1.5 dB gains on 3 linear inverse scattering tasks.

Code: <https://github.com/szhan311/lmaps>

2. Background

Unconditional diffusion models. The goal of diffusion model is to sample from an unknown distribution $\pi_0(x_0)$ given a training dataset $\mathcal{D} = \{x_0^i\}_{i=1}^N$. Given a data point $x_0 \sim \pi_0$ and a time step t , a noisy datapoint is sampled from the transition kernel: $p_t(x_t | x_0) = \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 \mathbb{I})$. Diffusion process is built by mixture of densities: $p_t(x_t) = \int p_t(x_t | x_0) \pi_0(x_0) dx_0$, and DDIM samples $\pi_0(x_0)$ by running an iterative process $p_t(x_t)$ from time $t = T$ to $t = 0$ with the initial condition $x_T \sim p(x_T)$:

$$x_{t-\Delta t} = g(m_{0|t}(x_t), x_t, \epsilon), \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}) \quad (1)$$

where $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ is the fresh noise added at the inference time, $m_{0|t}(t, x) = \mathbb{E}[x_0 | x_t]$ is the ideal denoiser, and we define:

$$g(\xi, x_t, \epsilon) := \alpha_{t-\Delta t} \xi + \sigma_{t-\Delta t} \left(\sqrt{1 - \rho_t^2} \frac{x_t - \alpha_t \xi}{\sigma_t} + \rho_t \epsilon \right), \quad (2)$$

The goal of posterior sampling is to generate samples under some condition y , i.e., sample x_0 from a posterior distribution, $\pi_{0|y}(x_0 | y)$, where y could be class labels, measurements or text information, for example. In this paper, we focus on two representative lines of posterior sampling approaches with diffusion priors: (i) the family of diffusion posterior sampling (DPS) methods based on Tweedie’s formula, and (ii) Decoupled Annealing Posterior Sampling (DAPS).

Diffusion Posterior Sampling (DPS) family. DPS generate $x_0 \sim \pi_{0|y}(x_0 | y)$ by running an iterative process $p_{t|y}(x_t | y)$ from time $t = T$ to $t = 0$ with the initial condition $x_T \sim p(x_T | y)$:

$$x_{t-\Delta t} = g(m_{0|t,y}(t, x_t, y), x_t, \epsilon), \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}), \quad (3)$$

where $m_{0|t,y}(t, x_t, y) = \mathbb{E}[x_0 | x_t, y]$ is the conditional denoiser. According to Tweedie’s formula,

$$\mathbb{E}[x_0 | x_t, y] = m_{0|t} + \frac{\sigma_t^2}{\alpha_t} \nabla_{x_t} \log p(y | x_t). \quad (4)$$

Equation (4) connects the conditional denoiser $\mathbb{E}[x_0 | x_t, y]$ with the unconditional denoiser $\mathbb{E}[x_0 | x_t]$. However, the additional term $\nabla_{x_t} \log p(y | x_t)$ is still intractable. One can train a neural network to approximate $\nabla_{x_t} \log p(y | x_t)$, like classifier guidance (Dhariwal & Nichol, 2021). Training-free guidance, such as in (Chung et al., 2022), usually approximates $\nabla_{x_t} \log p(y | x_t)$ by a convenient single-sample approximation, $p(y | x_t) \approx p(y | m_{0|t}(x_t))$, according to chain rule:

$$\nabla_{x_t} \log p(y | x_t) \approx \nabla_{x_t} m_{0|t}(t, x_t) \nabla_{m_{0|t}} \log p(y | m_{0|t}). \quad (5)$$

Decoupled Annealing Posterior Sampling (DAPS) (Zhang et al., 2025a). Alternatively, DAPS developed a new framework to sample $x_0 \sim \pi_{0|y}(x_0 | y)$, which is given by the following iterations:

$$\begin{aligned} x_{0|t,y} &\sim p(x_0 | x_t, y) \\ x_{t-\Delta t} &\sim \mathcal{N}(\alpha_{t-\Delta t} x_0, \sigma_{t-\Delta t}^2 \mathbb{I}). \end{aligned} \quad (6)$$

Approximate posterior samples $x_{0|t,y}$ are obtained at each diffusion step using Langevin dynamics.

DMAP (Xu et al., 2025). DMAP provides a closely related MAP-based perspective on diffusion inverse-problem solvers. Instead of drawing posterior samples, DMAP formulates a local MAP update in the latent diffusion variable, e.g., by optimizing a mode of $p(x_{t-\Delta t} | x_t, y)$ at each reverse step, using denoiser-based approximations to connect latent variables with the clean signal and measurement likelihood. This makes DMAP an important MAP-oriented counterpart to DPS/DAPS-style posterior sampling methods and a close point of comparison for LMAPS.

3. Local MAP Sampling

3.1. Local MAP and global MAP

Global MAP. In Bayesian inference, the maximum a posteriori (MAP) estimate is defined as the single configuration

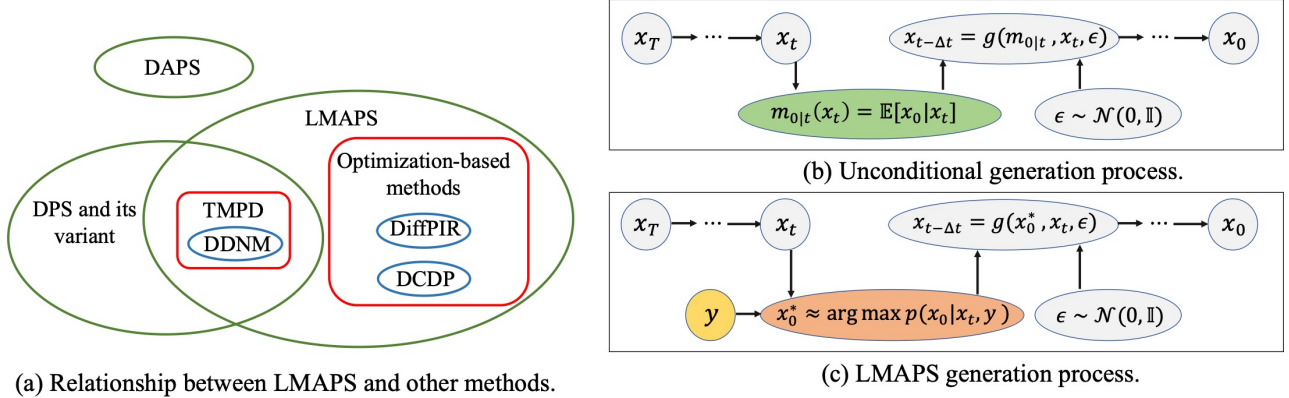


Figure 1. Comparison of LMAPS with other methods. (a). The relationship between different alignment approaches; (b). The generation process of unconditional diffusion model; (c). The generation process of LMAPS.

Algorithm 1 DPS	Algorithm 2 DAPS	Algorithm 3 LMAPS
1: Input: $x_{t_N} \sim \pi_T$ 2: for $k = N$ to 1 do 3: $\tilde{x}_0 = \mathbb{E}[x_0 x_{t_k}, y]$ 4: $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ 5: $x_{t_{k-1}} = g(\tilde{x}_0, x_{t_k}, \epsilon)$ 6: end for 7: return x_0	1: Input: $x_{t_N} \sim \pi_T$ 2: for $k = N$ to 1 do 3: $\tilde{x}_0 \sim p(x_0 x_{t_k}, y)$ 4: $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ 5: $x_{t_{k-1}} \sim \mathcal{N}(\alpha_{t_{k-1}} x_0, \sigma_{t_{k-1}}^2 \mathbb{I})$ 6: end for 7: return x_0	1: Input: $x_{t_N} \sim \pi_T$ 2: for $k = N$ to 1 do 3: $\tilde{x}_0 = \arg \max p(x_0 x_{t_k}, y)$ 4: $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ 5: $x_{t_{k-1}} = g(\tilde{x}_0, x_{t_k}, \epsilon)$ 6: end for 7: return x_0

Figure 2. Comparison of inference algorithm between DPS, DAPS and LMAPS.

that maximizes the posterior probability,

$$x_0^{\text{MAP}} := \arg \max_{x_0} p(x_0 | y). \quad (7)$$

We refer to this as the *global MAP*, since it directly targets the mode of the full posterior distribution after conditioning on the observation y . Unlike posterior sampling methods (e.g., DPS or DAPS), which produce diverse draws from $p(x_0 | y)$, global MAP yields a point estimate corresponding to (one of) the maximizers of the posterior. This estimate prioritizes fidelity and certainty over diversity, offering a principled way to recover a solution that best aligns with both the diffusion prior and the measurement model.

Local MAP. Directly solving for x_0^{MAP} in high-dimensional, non-convex posteriors can be computationally intractable. Instead, we consider a sequence of *local MAP* problems, which implemented by DDIM-like iteration from time $t = T$ to $t = 0$ with the initial condition $x_T \sim p(x_T | y)$:

$$x_0^*(t, x_t, y) := \arg \max_{x_0} p(x_0 | x_t, y), \quad (8a)$$

$$x_{t-\Delta t} = g(x_0^*, x_t, \epsilon), \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}). \quad (8b)$$

Equation (8a) and Equation (8b) correspond to the local MAP step and the DDIM update step, respectively. In particular, the local MAP step is equivalent to:

$$x_0^* = \arg \min \{-\log p(x_0 | x_t) - \log p(y | x_0)\}. \quad (9)$$

This optimization problem can be solved via gradient descent if $\log p(x_0 | x_t)$ and $\log p(y | x_0)$ are known and differentiable, although in practice we approximate $p(x_0 | x_t)$ as discussed in Section 4.

Relation to DMAP. DMAP (Xu et al., 2025) is the closest related formulation and shares the same high-level motivation of using a local MAP perspective along the diffusion trajectory. The main modeling difference is the variable on which the local MAP problem is posed. DMAP considers a latent-space mode of the form $x_{t-\Delta t}^* = \arg \max p(x_{t-\Delta t} | x_t, y)$, whereas LMAPS solves the clean-space local MAP problem $x_0^* = \arg \max p(x_0 | x_t, y)$ and then applies the diffusion transition in Equation (8b). In a deterministic ODE/DDIM limit, these viewpoints are closely connected because the clean-space estimate is transported through a deterministic transition. With stochastic transitions, the latent-space objective additionally includes the effect of injected transition noise, while the clean-space formulation keeps the data-consistency step directly tied to $p(y | x_0)$.

3.2. The difference between DPS, local MAP and global MAP

One might expect that the iteration in Equation (8) can be used to sample from the posterior $p(x_0 | y)$ or converge to global MAP $\arg \max p(x_0 | y)$. Unfortunately, this is

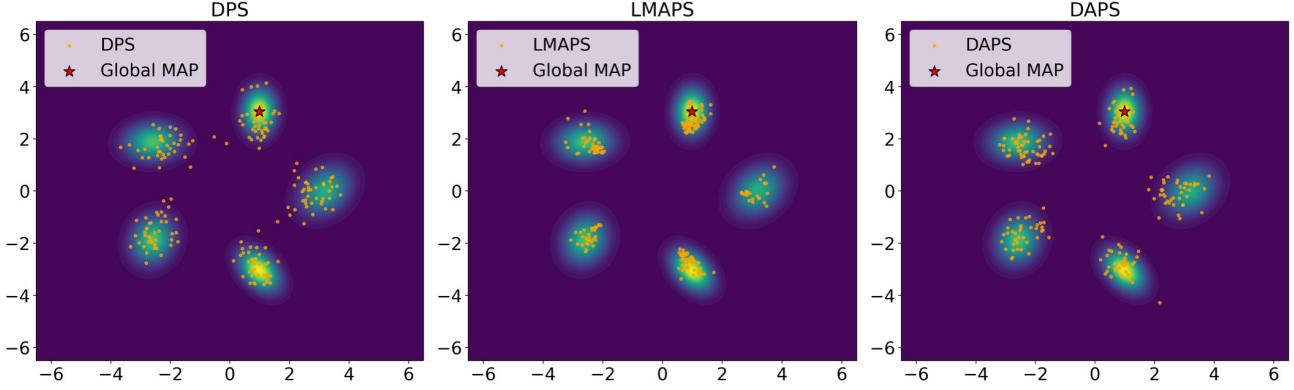


Figure 3. Comparison of LMAPS, DPS, DAPS and Global MAP on 2D synthetic data, here we assume $p(x_0 | y)$ is a Gaussian mixture which have analytical expression (see Appendix A). LMAPS is less likely to generate samples in the between-mode regions or low-density regions.

generally not the case.

DPS vs. local MAP. DPS evolves x_t by using the conditional mean $m_{0|t,y}(t, x_t, y) = \mathbb{E}[x_0 | x_t, y]$ inside the DDIM update (Equation (3)), whereas local MAP replaces the mean with the conditional mode: $x_0^*(t, x_t, y) = \arg \max p(x_0 | x_t, y)$, and then plugs x_0^* into the same $g(\cdot)$ transition (Equation (8)). Consequently, replacing $\mathbb{E}[x_0 | x_t, y]$ with $\arg \max p(x_0 | x_t, y)$ alters the forward operator acting on $p_{t|y}(x_t)$ and does not preserve the posterior marginals $p_{t|y}$.

When are DPS and local MAP equivalent? These two coincide if and only if $\mathbb{E}[x_0 | x_t, y] = \arg \max p(x_0 | x_t, y)$, for example if $p(x_0 | x_t, y)$ is (uni-variate or multi-variate) Gaussian. The condition holds, e.g., in linear-Gaussian inverse problems with a Gaussian diffusion prior approximation (quadratic negative log-density), with detailed discussion in Section 4. Outside of this setting (nonlinear forward models, heavy-tailed likelihoods, mixture-like priors), the posterior $p(x_0 | x_t, y)$ is non-Gaussian and the two updates generally differ. With non-Gaussian $p(x_0 | x_t, y)$, local MAP introduces a mode-seeking bias and does not reproduce posterior sampling.

Local MAP vs. global MAP. a global MAP solution is any maximizer of $x_0^{\text{MAP}} = \arg \max p(x_0 | y)$. Local MAP instead solves, at each time t , a conditioned optimization (Equation (9)): $x_0^*(t, x_t, y) = \arg \max p(x_0 | x_t, y)$. Because x_t itself depends on the entire past trajectory (initialization, noise schedule, and random seeds), the sequence of local maximizers need not approach the global maximizer of $p(x_0 | y)$ as $t \downarrow 0$.

In summary, DPS targets $p(x_0 | y)$, and LMAPS targets $\arg \max p(x_0 | x_t, y)$ at each step. Local MAP equals DPS only in Gaussian conditional settings; outside them, local MAP generally does not sample the posterior and can fail to reach the global MAP. We visualize a toy example in

Figure 3. Compared to DPS and DAPS, LMAPS is less likely to generate samples in between-mode regions or low-density regions.

4. Local MAP sampling for inverse problem

The primary goal of solving an inverse problem is to recover an unknown image or signal $x_0 \in \mathbb{R}^n$ from a prior distribution, $\pi(x_0)$, and noisy measurement $y \in \mathbb{R}^m$. Mathematically, the unknown signal and the measurements are related by a forward model:

$$y = \mathcal{H}(x_0) + z \quad (10)$$

where $\mathcal{H}(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (with $m < n$) represents the linear or non-linear forward operator, $z \in \mathbb{R}^m$ denotes the noise in the measurement domain. We assume the added noise z is sampled from a Gaussian distribution $\mathcal{N}(0, \sigma_y^2 \mathbb{I})$, where $\sigma_y > 0$ denotes the noise level. The forward operator and Equation (10) define the likelihood $p(y | x_0)$ for both the global or local MAP problems in Section 3.1.

The final ingredient for constructing a local posterior and solving the resulting MAP problem is the choice of prior $p(x_0 | x_t)$. While the true transition kernel of a diffusion model prior requires simulation, we can proceed as in previous work (Boys et al., 2023; Song et al., 2023b) by projecting onto the first two moments using a Gaussian approximation, $p(x_0 | x_t) \approx \mathcal{N}(x_0; m_{0|t}, \Sigma_{0|t})$, where $m_{0|t}(x_t) := \mathbb{E}[x_0 | x_t]$. While Boys et al. (2023) show that $\Sigma_{0|t}^{\text{TPD}}(x_t) := \mathbb{E}[(x_0 - m_{0|t})(x_0 - m_{0|t})^T | x_t] = \frac{\sigma_t^2}{\alpha_t} \nabla_{x_t} m_{0|t}$, we will consider flexible choices of $\Sigma_{0|t}$. Finally, the local MAP problem amounts to solving

$$x_0^* = \arg \min_{x_0} \left\{ (x_0 - m_{0|t})^\top \Sigma_{0|t}^{-1} (x_0 - m_{0|t}) + \frac{1}{\sigma_y^2} \|y - \mathcal{H}(x_0)\|^2 \right\}. \quad (11)$$

We will develop methodology for approximately solving the local MAP problem for general non-linear inverse problems in Section 4.1, before discussing the case of linear inverse problems in Section 4.2.

4.1. Approximated solution for nonlinear inverse problems

Isotropic approximation of $\Sigma_{0|t}$. For nonlinear $\mathcal{H}(\cdot)$, there is no explicit solution for x_0^* and it would be more expensive to adopt the moment projection covariance $\Sigma_{0|t} = \nabla_{0|t}^{\text{TMPD}} = \frac{\sigma_t^2}{\alpha_t} \nabla_{x_t} m_{0|t}$.

For a Gaussian prior $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$, the exact posterior covariance under the forward noising process $x_t = \alpha_t x_0 + \sigma_t \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ is

$$\Sigma_{0|t} = (\Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I})^{-1} = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I} + \mathcal{O}\left(\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2\right) \preceq \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I}, \quad (12)$$

so the leading term is isotropic and all anisotropy appears only as higher-order corrections as $t \rightarrow 0$ (i.e., $\sigma_t^2 \rightarrow 0$ and $\alpha_t \rightarrow 1$). More generally, even for non-Gaussian priors $p(x_0)$ with a smooth log-density, the Hessian satisfies

$$\nabla_{x_0}^2 [-\log p(x_0 | x_t)] = \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} + \nabla_{x_0}^2 [-\log p(x_0)]. \quad (13)$$

As $\sigma_t^2 \rightarrow 0$, the isotropic data term $\frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}$ dominates the prior curvature, implying that the local Gaussian approximation to $p(x_0 | x_t)$ is asymptotically isotropic. We provide a formal statement and proof in Appendix B. Motivated by the above analysis, we approximate the conditional covariance by an isotropic form $\Sigma_{0|t} \approx \frac{1}{\text{SNR}} \mathbb{I}$, where $\text{SNR} := \alpha_t^2 / \sigma_t^2$. This approximation captures the leading-order behavior of the true posterior covariance as $t \rightarrow 0$. In practice, we further introduce a tunable parameter k that adjusts the relative influence between the denoising estimate $m_{0|t}$ and the measurement y . Equivalently, introducing k can be viewed as optimizing a tempered local posterior, where the local diffusion prior is raised to a temperature-dependent power; see Appendix C for a detailed discussion. With this modification, the MAP objective becomes

$$x_0^* = \arg \min_{x_0} \left\{ \frac{\text{SNR}}{k} \|x_0 - m_{0|t}\|^2 + \frac{1}{\sigma_y^2} \|y - \mathcal{H}(x_0)\|^2 \right\}. \quad (14)$$

Objective Reformulation. In the implementation, the weighting of the two terms in Equation (14) depends on raw signal-to-noise ratios, which can vary drastically with t , which makes it difficult to choose the appropriate learning rate. For analysis and implementation it is convenient to reformulate Equation (14) in a scale-invariant way.

Multiplying the objective by a positive constant (which does not change the minimizer) and introducing parameters $k_1, k_2 > 0$ such that $2k_2/k_1^2 = k/(\alpha_t^2 \sigma_y^2)$, we obtain the equivalent problem

$$x_0^* = \arg \min_{x_0} \left\{ \left(1 - \frac{\sigma_t^2}{\sigma_t^2 + k_1^2}\right) \frac{1}{2} \|x_0 - m_{0|t}\|^2 + \frac{\sigma_t^2}{\sigma_t^2 + k_1^2} k_2 \|y - \mathcal{H}(x_0)\|^2 \right\}. \quad (15)$$

This reformulation has several advantages:

- **Convex-combination interpretation.** The weights can be written as $(1 - \mu_t)$ and μ_t with $\mu_t = \sigma_t^2 / (\sigma_t^2 + k_1^2) \in (0, 1)$. Thus the cost is a convex combination of the prior and data fidelity terms.
- **Automatic annealing.** As σ_t^2 decreases over time, μ_t gradually shifts the objective from measurement-driven ($\mu_t \approx 1$) to prior-driven ($\mu_t \approx 0$).
- **Interpretable parameters.** The scale k_1 plays the role of a trust-region parameter balancing prior and measurement, while k_2 is a scale factor for the consistency loss to the measurement.
- **Numerical stability.** Keep weights in $[0, 1]$ avoids extreme scaling from SNR values, improving conditioning and optimizer robustness.

In the implementation, we adopt gradient descent to solve x_0^* in Equation (15), the algorithm of LMAPS for inverse problems is provided in Algorithm 4.

Relationship to optimization-based methods. Previous optimization-based approaches (Song et al., 2023a; Li et al., 2024; Zhu et al., 2023) solve for x_0^* through the following objective:

$$x_0^* = \arg \min \|x_0 - m_{0|t}\|^2 + \lambda_t \|y - \mathcal{H}(x_0)\|^2, \quad (16)$$

where λ_t is a hyperparameter, often chosen heuristically without a principled basis. These methods can be viewed as special cases of our framework by setting $\Sigma_{0|t} = \lambda_t \sigma_y^2 \mathbb{I}$ in Equation (11).

While the objectives in Equation (16) and Equation (15) are indeed equivalent, we found that empirical performance strongly depends on our objective reformulation and choices of weighting terms as motivated above. Further, our local MAP interpretation provides a probabilistic perspective for these objectives and suggests the connection with TMPD in the case of linear inverse problems, as discussed in Section 4.2.

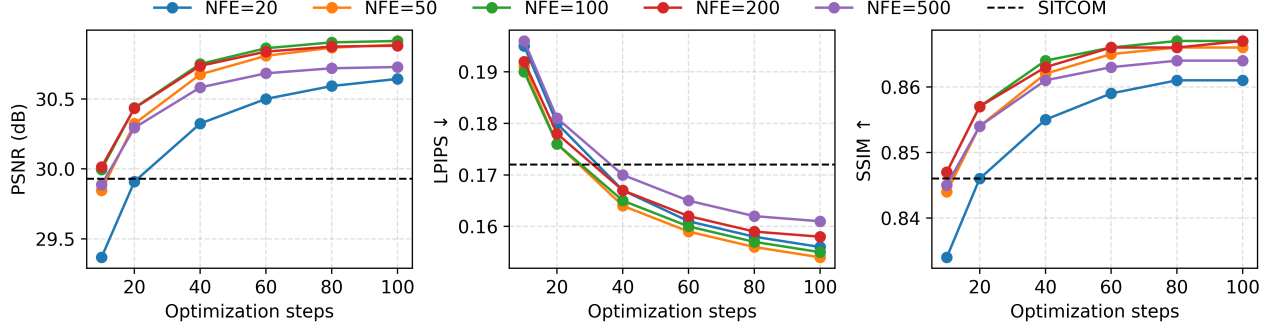


Figure 4. Ablation study on optimization steps vs. diffusion steps (NFEs) for Gaussian Deblurring.

Algorithm 4 Local MAP Sampling (LMAPS) for inverse problems.

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1: Input: measurement  $y$ ; forward operator  $\mathcal{H}(\cdot)$ ; pretrained DM  $\epsilon_\theta(\cdot)$ ;
2:   diffusion steps  $N$ ; schedule  $\{\alpha_n, \sigma_n\}_{n=1}^N$ ; gradient updates  $K$ ;
3:   objective params  $k_1, k_2$ ; step size  $\eta$ .
4: Initialize:  $x_N \sim \mathcal{N}(0, \mathbf{I})$ .
5: for  $n = N$  down to 1 do
6:    $\hat{x}_0 \leftarrow (x_n - \sigma_n \epsilon_\theta(x_n, n)) / \alpha_n$ 
7:   // predicted clean sample
8:    $r \leftarrow \sigma_n^2 / (\sigma_n^2 + k_1^2 + 10^{-6})$ 
9:    $x'_0 \leftarrow \hat{x}_0$ 
10:  for  $k = 1$  to  $K$  do
11:     $g \leftarrow (1 - r)(x'_0 - \hat{x}_0) + r k_2 \nabla_{x'_0} \|y - \mathcal{H}(x'_0)\|^2$ 
12:    // gradient of Equation (15)
13:     $x'_0 \leftarrow x'_0 - \eta g$ 
14:  end for
15:   $x_{n-1} \sim \mathcal{N}(\alpha_{n-1} x'_0, \sigma_{n-1}^2 \mathbf{I})$ 
16:  // diffusion transition
17: end for
18: Output:  $x'_0$ 
    
```

4.2. Exact Solution for Linear Inverse Problems

As discussed in Section 3.2, the *local MAP solution matches the posterior mean* for Gaussian posteriors $p(x_t | x_0, y)$ arising from linear inverse problems $p(y | x_0) = \mathcal{N}(Hx_0, \sigma_y^2 \mathbf{I})$ with a Gaussian assumption on the prior $p(x_0 | x_t) = \mathcal{N}(x_0; m_{0|t}, \Sigma_{0|t})$. Solving in closed form for the posterior mean as in (Boys et al., 2023), we have

$$\begin{aligned}
 x_0^* &= m_{0|t} + \Sigma_{0|t} H^T (H \Sigma_{0|t} H^T + \sigma_y^2 \mathbf{I})^{-1} (y - H m_{0|t}). \\
 &= m_{0|t} + \frac{\sigma_t^2}{\alpha_t} \nabla_{x_t} \log p(y | x_t)
 \end{aligned} \tag{17}$$

We recover Tweedie Moment-Projected Diffusion (Boys et al., 2023) as a special case for $\Sigma_{0|t}^{\text{TMPD}} = \frac{\sigma_t^2}{\alpha_t} \nabla_{x_t} m_{0|t}(x_t)$, which is expensive since it requires the gradient with respect to the denoiser $m_{0|t}$. Thus, Local MAP Sampling reduced

to DPS.

When applying LMAPS to linear inverse problems, we assume $\Sigma_{0|t} = \frac{k}{\text{SNR}_t} \mathbf{I}$ as in Section 4.1, and optimize with K steps of gradient descent at each timestep despite the availability of the closed form in Equation (17). We include solving LMAPS with analytical solution in Appendix F.3.

5. Experiments

5.1. Experimental setup

Inverse problems. We evaluate our method on image restoration and scientific inverse problems. For linear image restoration, we consider (1) super-resolution, (2) Gaussian deblurring, (3) motion deblurring, (4) inpainting (with a box mask), and (5) inpainting (with a 70% random mask). For nonlinear image restoration, we consider (1) phase retrieval, (2) high dynamic range (HDR) reconstruction, (3) nonlinear deblurring, (4) JPEG restoration, (5) quantization, where HDR, JPEG restoration and quantization are nonlinear inverse problems with non-differentiable operators. For scientific inverse problems, we adopt the benchmark from InverseBench (Zheng et al., 2025), which includes Linear Inverse Scattering (LIS), Compressed sensing MRI (CS-MRI) and Black Hole Imaging. More details are provided in Appendix E.

Dataset and Pretrained models. For image restoration, we evaluated our method on FFHQ (Karras et al., 2019) 256×256 and ImageNet 256×256 datasets (Deng et al., 2009). Following DAPS, we test the same subset of 100 images for both datasets. For scientific inverse problems, we adopt the same dataset as InverseBench (Zheng et al., 2025). For image restoration tasks, we utilize the pre-trained checkpoint (Chung et al., 2022) on the FFHQ dataset and the pre-trained checkpoint (Dhariwal & Nichol, 2021) on the ImageNet dataset. For scientific inverse problems, we adopt the pre-trained checkpoints from InverseBench.

Baselines. We compare our method with the following baselines: DDNM (Wang et al., 2022), DDRM (Kawar

Table 1. Quantitative evaluation of solving image restoration FFHQ (left) and ImageNet (right), with Gaussian noise ($\sigma_y = 0.05$): 5 linear and 5 nonlinear tasks (3 non-differentiable). Results are reported as mean PSNR, SSIM, LPIPS, and FID across 100 images. Best results are highlighted in bold. For phase retrieval, DAPS and LMAPS select the best result from 4 runs for each image.

Task	Method	FFHQ				ImageNet			
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
SR 4 $\times$	DPS	25.86	0.753	0.269	81.70	21.13	0.489	0.361	106.32
	DDRM	26.58	0.782	0.282	79.25	22.62	0.521	0.324	103.85
	DDNM	28.03	0.795	0.197	64.62	23.96	0.604	0.475	98.62
	DCDP	28.66	0.807	0.178	53.81	—	—	—	—
	FPS-SMC	28.42	0.813	0.204	49.25	24.82	0.703	0.313	97.51
	DiffPIR	26.64	—	0.260	65.77	23.18	—	0.371	106.32
	DAPS	29.07	0.818	0.177	51.44	25.89	0.694	0.276	83.57
	DMPlug	28.86	0.820	0.128	73.72	—	—	—	—
	MMPS	28.45	0.811	0.106	54.05	—	—	—	—
	SITCOM	30.55	0.864	0.154	62.70	27.07	0.746	0.228	89.71
	MGDM	27.81	0.798	0.111	45.96	25.44	0.684	0.246	77.72
	MAP-GA	29.97	0.844	0.178	81.63	26.00	0.708	0.267	138.30
	DMAPI	28.56	0.783	0.165	44.78	25.39	0.661	0.229	74.65
	LMAPS	30.74	0.869	0.165	80.63	26.72	0.739	0.242	121.27
Inpaint (Box)	DPS	22.51	0.792	0.209	61.27	18.94	0.722	0.257	126.52
	DDRM	22.26	0.801	0.207	78.62	18.63	0.733	0.254	116.37
	DDNM	24.47	0.837	0.235	46.59	21.64	0.748	0.319	103.97
	DCDP	23.89	0.760	0.163	45.23	—	—	—	—
	FPS-SMC	24.86	0.823	0.146	48.34	22.16	0.726	0.208	111.58
	DAPS	24.07	0.814	0.133	43.10	21.43	0.725	0.214	109.85
	SITCOM	24.95	0.849	0.131	64.83	19.72	0.784	0.164	102.91
	MMPS	23.38	0.853	0.084	39.29	—	—	—	—
	MAP-GA	24.77	0.850	0.123	48.82	20.71	0.802	0.198	146.02
	DMAPI	22.37	0.817	0.136	78.40	18.71	0.757	0.188	134.94
	LMAPS	25.02	0.876	0.108	45.03	21.25	0.803	0.204	146.68
Inpaint (Random)	DPS	25.46	0.823	0.203	69.20	23.52	0.745	0.297	87.53
	DDNM	29.91	0.817	0.121	44.37	31.16	0.841	0.191	63.84
	DCDP	30.69	0.842	0.142	52.51	—	—	—	—
	FPS-SMC	28.21	0.823	0.261	61.23	24.52	0.701	0.316	79.12
	DAPS	31.12	0.844	0.098	32.17	28.44	0.775	0.135	54.25
	SITCOM	33.96	0.928	0.082	31.23	29.74	0.855	0.115	30.25
	DMPlug	31.55	0.892	0.110	72.69	—	—	—	—
	MMPS	31.91	0.905	0.041	28.15	—	—	—	—
	MAP-GA	32.00	0.908	0.088	39.14	28.09	0.830	0.143	61.67
	DMAPI	32.43	0.886	0.105	29.87	28.78	0.816	0.139	42.52
	LMAPS	34.51	0.938	0.066	28.60	30.59	0.876	0.100	33.53
Gaussian Deblurring	DPS	25.87	0.764	0.219	79.75	20.31	0.598	0.397	116.42
	DDRM	24.93	0.732	0.239	92.43	21.26	0.564	0.443	146.89
	DCDP	27.50	0.699	0.304	86.43	—	—	—	—
	FPS-SMC	26.54	0.773	0.253	67.45	23.91	0.601	0.387	91.72
	DiffPIR	27.36	—	0.236	59.65	22.80	—	0.355	93.36
	DAPS	29.19	0.817	0.165	53.33	26.15	0.684	0.253	75.68
	SITCOM	29.93	0.846	0.172	73.24	26.39	0.716	0.260	110.95
	MGDM	27.78	0.791	0.110	43.31	25.50	0.682	0.289	93.84
	DMAPI	26.83	0.745	0.181	53.97	23.80	0.578	0.267	96.15
	LMAPS	30.88	0.867	0.158	83.78	26.65	0.727	0.250	133.13
Motion Deblurring	DPS	24.52	0.801	0.246	65.23	18.96	0.629	0.423	137.81
	DCDP	25.08	0.512	0.364	125.13	—	—	—	—
	FPS-SMC	27.39	0.826	0.227	48.32	24.52	0.647	0.326	87.43
	DiffPIR	26.57	—	0.255	65.78	24.01	—	0.366	94.63
	DAPS	29.66	0.847	0.157	39.49	27.86	0.766	0.196	61.83
	SITCOM	29.36	0.840	0.185	70.52	26.76	0.746	0.242	105.15
	MMPS	31.15	0.870	0.075	40.74	—	—	—	—
	MGDM	26.72	0.776	0.124	49.45	24.52	0.659	0.278	102.09
	DMAPI	27.52	0.762	0.194	55.83	22.22	0.571	0.338	136.23
	LMAPS	32.62	0.902	0.117	54.79	28.42	0.796	0.204	86.63
Phase Retrieval	DPS	17.64 $\pm$ 2.97	0.441 $\pm$ 0.129	0.410 $\pm$ 0.090	104.52	16.81 $\pm$ 3.61	0.427 $\pm$ 0.143	0.447 $\pm$ 0.099	197.54
	RED-diff	15.60 $\pm$ 4.48	0.398 $\pm$ 0.195	0.596 $\pm$ 0.092	167.43	14.98 $\pm$ 3.75	0.386 $\pm$ 0.057	0.536 $\pm$ 0.129	212.24
	MGDM	19.24 $\pm$ 8.22	0.533 $\pm$ 0.271	0.346 $\pm$ 0.254	157.28	13.77 $\pm$ 4.30	0.293 $\pm$ 0.196	0.578 $\pm$ 0.169	279.06
	DAPS	30.63 $\pm$ 3.13	0.851 $\pm$ 0.072	0.139 $\pm$ 0.060	42.71	21.39 $\pm$ 6.59	0.473 $\pm$ 0.226	0.372 $\pm$ 0.166	82.67
	LMAPS	31.56$\pm$3.02	0.867$\pm$0.057	0.126$\pm$0.052	34.80	22.86$\pm$7.50	0.596$\pm$0.267	0.313$\pm$0.176	133.22
Nonlinear Deblurring	DPS	23.39 $\pm$ 2.01	0.263 $\pm$ 0.082	0.278 $\pm$ 0.060	91.31	22.49 $\pm$ 3.20	0.591 $\pm$ 0.101	0.306 $\pm$ 0.081	101.41
	RED-diff	30.86$\pm$0.51	0.795 $\pm$ 0.028	0.160 $\pm$ 0.034	43.84	30.07$\pm$1.41	0.754 $\pm$ 0.023	0.211 $\pm$ 0.083	51.22
	DCDP	27.92 $\pm$ 2.64	0.779 $\pm$ 0.067	0.183 $\pm$ 0.051	51.96	—	—	—	—
	DAPS	28.29 $\pm$ 1.77	0.783 $\pm$ 0.036	0.155 $\pm$ 0.032	49.38	27.73 $\pm$ 3.23	0.724 $\pm$ 0.048	0.169 $\pm$ 0.056	59.87
	DMPlug	27.65 $\pm$ 2.98	0.799 $\pm$ 0.080	0.181 $\pm$ 0.056	92.64	—	—	—	—
	SITCOM	29.19 $\pm$ 2.35	0.789 $\pm$ 0.093	0.190 $\pm$ 0.014	51.26	28.55 $\pm$ 3.87	0.798$\pm$0.092	0.149$\pm$0.050	44.67
	MGDM	23.88 $\pm$ 2.61	0.664 $\pm$ 0.081	0.271 $\pm$ 0.085	101.04	22.63 $\pm$ 2.98	0.583 $\pm$ 0.122	0.394 $\pm$ 0.117	192.29
High Dynamic Range	LMAPS	29.93 $\pm$ 1.83	0.855$\pm$0.035	0.150$\pm$0.034	57.28	28.03 $\pm$ 3.62	0.774 $\pm$ 0.099	0.183 $\pm$ 0.065	81.21
	DPS	22.73 $\pm$ 6.07	0.591 $\pm$ 0.141	0.264 $\pm$ 0.156	112.82	19.23 $\pm$ 2.52	0.582 $\pm$ 0.082	0.503 $\pm$ 0.106	146.23
	DAPS	27.12 $\pm$ 3.53	0.752 $\pm$ 0.041	0.162 $\pm$ 0.072	42.97	26.30 $\pm$ 4.10	0.717 $\pm$ 0.067	0.175 $\pm$ 0.107	64.19
	SITCOM	28.02 $\pm$ 3.28	0.812 $\pm$ 0.108	0.174 $\pm$ 0.081	55.65	25.59 $\pm$ 3.66	0.170 $\pm$ 0.141	0.198 $\pm$ 0.177	64.52
	MGDM	25.73 $\pm$ 4.28	0.796 $\pm$ 0.151	0.100$\pm$0.096	42.33	23.43 $\pm$ 4.68	0.754 $\pm$ 0.165	0.173 $\pm$ 0.152	67.83
JPEG Restoration (QF=5)	LMAPS	28.87$\pm$3.39	0.884$\pm$0.082	0.141 $\pm$ 0.074	44.25	27.02$\pm$4.00	0.860$\pm$0.096	0.158$\pm$0.090	46.53
	MGDM	25.04 $\pm$ 1.28	0.755 $\pm$ 0.060	0.270 $\pm$ 0.045	78.00	22.41 $\pm$ 2.23	0.606 $\pm$ 0.144	0.417 $\pm$ 0.087	185.66
Quantization	LMAPS	27.25$\pm$1.37	0.814$\pm$0.045	0.260$\pm$0.043	110.07	24.96$\pm$2.46	0.703$\pm$0.124	0.340$\pm$0.089	164.43
	MGDM	25.82 $\pm$ 1.29	0.789 $\pm$ 0.063	0.255 $\pm$ 0.046	85.86	22.34 $\pm$ 2.26	0.425 $\pm$ 0.110	0.605 $\pm$ 0.156	198.19
	LMAPS	29.51$\pm$1.14	0.844$\pm$0.467	0.229$\pm$0.474	104.67	26.92$\pm$2.25	0.748$\pm$0.114	0.307$\pm$0.099	150.87

Table 2. Quantitative evaluation of solving scientific inverse problems is conducted using PSNR as the evaluation metric. The tasks include: (i) three LIS settings with different numbers of receivers (NR = 360, 180, 60); (ii) four CS-MRI settings with varying subsampling ratios (4 $\times$, 8 $\times$) and measurement types (noiseless and raw); and (iii) three Black Hole Imaging settings with different observation time ratios (3%, 10%, 100%).

Method	LIS			CS-MRI				Black Hole		
	NR=360	NR=180	NR=60	4 $\times$ noiseless	4 $\times$ raw	8 $\times$ noiseless	8 $\times$ raw	100%	10%	3%
DDRM	32.13	28.08	20.44	—	—	—	—	—	—	—
DDNM	26.28	35.02	29.24	—	—	—	—	—	—	—
IIGDM	27.93	26.40	20.07	—	—	—	—	—	—	—
DPS	32.06	31.80	27.37	26.13	25.83	20.82	23.00	25.86	24.36	24.20
LGD	27.90	27.84	20.49	—	—	—	—	21.22	22.08	22.51
DiffPIR	34.24	34.01	26.32	28.31	27.60	26.78	26.26	25.01	23.84	24.12
PnP-DM	33.94	31.82	24.72	31.80	27.62	29.33	25.28	26.07	24.57	24.25
DAPS	34.64	33.16	25.88	31.48	28.61	29.01	27.10	25.60	23.99	23.54
RED-diff	36.56	35.41	27.07	29.36	28.71	26.76	27.33	23.77	22.53	20.74
FPS	33.24	29.62	21.32	—	—	—	—	—	—	—
MCG-diff	30.94	28.06	21.00	—	—	—	—	—	—	—
LMAPS	38.07	37.19	30.75	32.83	28.77	30.50	27.43	26.79	24.83	24.66

et al., 2022), IIGDM (Song et al., 2023b), DPS (Chung et al., 2022), LGD (Song et al., 2023c), PnP-DM (Wu et al., 2024), FPS (Dou & Song, 2024), MCG-diff (Cardoso et al., 2023), RedDiff (Mardani et al., 2023), DAPS (Zhang et al., 2025a), DiffPIR (Zhu et al., 2023), DCDP (Li et al., 2024), SITCOM (Alkhouri et al., 2024), DMPlug (Wang et al., 2024), MGDM (Janati et al., 2025), MAP-GA (Gutha et al., 2025), MMPS (Rozet et al., 2024), DMAP (Xu et al., 2025).

Metrics. For image restoration tasks, we report Peak Signal-to-Noise Ratio (PSNR), Structural SIMilarity Index (SSIM), Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al., 2018), and Fréchet Inception Distance (FID) (Heusel et al., 2017). For scientific inverse problems, we primarily present PSNR in the main text, while additional task-specific metrics are provided in Appendix F.

5.2. Main results

Baselines. We compare our method with the following baselines: DDNM (Wang et al., 2022), DDRM (Kawar et al., 2022), IIGDM (Song et al., 2023b), DPS (Chung et al., 2022), LGD (Song et al., 2023c), PnP-DM (Wu et al., 2024), FPS (Dou & Song, 2024), MCG-diff (Cardoso et al., 2023), RedDiff (Mardani et al., 2023), DAPS (Zhang et al., 2025a), DiffPIR (Zhu et al., 2023), DCDP (Li et al., 2024), SITCOM (Alkhouri et al., 2024), DMPlug (Wang et al., 2024), MGDM (Janati et al., 2025), MAP-GA (Gutha et al., 2025), MMPS (Rozet et al., 2024), DMAP (Xu et al., 2025).

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5.3. Main results

Ablation studies. Figure 4 presents ablation studies on optimization steps across different diffusion steps. The best performance is typically observed at NFE = 200–500, where increasing the number of optimization steps per diffusion step yields notable improvements. Compared to the baseline SITCOM (600 NFEs with gradient computation through the U-Net), LMAPS attains similar performance while requiring substantially fewer computational resources. We report runtime comparisons for various methods on Deblurring task in Table 3 (Appendix D).

Image restoration. In Table 1, we present quantitative results for image restoration tasks on FFHQ and ImageNet datasets. The table covers 10 tasks, 4 restoration quality metrics, and 2 datasets, totaling 80 results. LMAPS achieves the best performance in 43 out of 60 cases for PSNR, SSIM, and LPIPS. Compared with the closely related DMAP baseline, LMAPS consistently improves distortion metrics on representative FFHQ tasks, including SR (30.74 vs. 28.56 PSNR), box inpainting (25.02 vs. 22.37), random inpainting (34.51 vs. 32.43), and Gaussian deblurring (30.88 vs. 26.83), with similar trends on ImageNet. The added FID evaluation is more nuanced: DMAP is better in some cases such as SR, while LMAPS is better in others such as inpainting and HDR, reflecting the trade-off between reconstruction fidelity and distribution-level similarity. Generally, LMAPS demonstrates superior performance than DAPS for most of the tasks with less computational cost. LMAPS improves > 2dB PSNR across motion deblurring, JPEG restoration and quantization tasks.

Scientific inverse problems. In Table 2, we report quantitative results of solving scientific inverse problems: Linear Inverse Scattering (LIS), CS-MRI, Black Hole Imaging. LMAPS demonstrates the best PSNR across all tasks, improved more than 1.5 dB PSNR for 3 LIS tasks.

6. Related work

Recent advances in conditional generation have led to breakthroughs in text-to-image synthesis, semantic editing, and domain-specific applications such as image-to-image translation and controlled signal reconstruction (Song et al., 2023c; Ye et al., 2024; Skreta et al., 2025; Singhal et al., 2025; Zheng et al., 2023). These methods have been especially impactful in solving inverse problems, including image restoration and scientific reconstruction tasks (Wang et al., 2022; Zheng et al., 2025). A wide range of approaches have been developed, spanning linear projection methods (Wang et al., 2022; Kavar et al., 2022; Zhang et al., 2025b; Dou & Song, 2024), Monte Carlo sampling (Wu et al., 2023; Phillips et al., 2024), variational inference (Feng et al., 2023; Mardani et al., 2023; Janati et al., 2024), and optimization-based strategies (Song et al., 2023a; Zhu et al., 2023; Li et al., 2024; Wang et al., 2024; Alkhouri et al., 2024; He et al., 2023).

Among these, Diffusion Posterior Sampling (DPS) and its variants (Zhang et al., 2025a; Chung et al., 2022; Song et al., 2023c; Yu et al., 2023; Rout et al., 2024; Yang et al., 2024; Bansal et al., 2023; Boys et al., 2023; Song et al., 2023b; Ho & Salimans, 2022) have gained wide adoption due to their strong empirical performance and interpretability, as they directly sample from the posterior distribution $p(x_0 | y)$. More recently, attention has shifted toward maximum a posteriori (MAP) estimation with diffusion priors. DMAP (Xu et al., 2025) is the closest related work to ours: it also interprets diffusion inverse-problem solvers through a MAP perspective and proposes local MAP-style updates along the diffusion trajectory. LMAPS follows this line of work, but poses the local MAP problem directly in clean-signal space, analyzes its relation to DPS, and connects the resulting objective to TMPD and optimization-based methods through the covariance approximation and objective reformulation. Finally, Gutha et al. (2025) proposed sampling from the global MAP solution, $\arg \max p(x_0 | y)$, though their approach is largely restricted to linear inverse problems.

7. Conclusion

We presented Local MAP Sampling (LMAPS), a new inference framework that iteratively solves local maximum-a-posteriori subproblems along the diffusion trajectory. By introducing a principled covariance approximation, an objective reformulation, and a gradient strategy for non-differentiable operators, LMAPS provides both theoretical clarity and practical effectiveness. Experiments across diverse image restoration and scientific inverse problems show that LMAPS consistently improves reconstruction quality, particularly on challenging tasks such as Box Inpainting, Phase Retrieval, JPEG restoration, and HDR.

Future work. In Bayesian inference, the global MAP plays a critical role and offers valuable insights contrasted with posterior sampling. Yet its utility has been relatively underexplored, and efficiently solving the global MAP with diffusion priors remains an open challenge. Advancing in this direction could enable more probable reconstructions and make contributions to solving inverse problems.

Impact Statement

This paper presents work whose primary goal is to advance the field of machine learning by improving the theoretical understanding and practical performance of diffusion-based inference methods. The techniques studied here are broadly applicable to inverse problems and share similar societal implications and ethical considerations with existing diffusion model research. We do not identify any specific negative societal impacts unique to this work.

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A. Gaussian mixture toy example

To gain intuition about posterior mean and MAP estimates in diffusion models, we consider a tractable toy prior $\pi_0(x_0)$ given by a Gaussian mixture:

$$\pi_0(x_0) = \sum_{k=1}^K \pi_k \mathcal{N}(x_0; \mu_k, \Sigma_k), \quad (18)$$

where $\pi_k > 0$ and $\sum_k \pi_k = 1$.

Forward kernel. As in the unconditional diffusion model, the forward corruption is

$$p_t(x_t | x_0) = \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 I). \quad (19)$$

Thus the marginal $p_t(x_t) = \int p_t(x_t | x_0) \pi_0(x_0) dx_0$ is itself a Gaussian mixture.

Posterior distribution. By Bayes' rule,

$$p(x_0 | x_t) \propto p_t(x_t | x_0) \pi_0(x_0). \quad (20)$$

Conditioned on mixture component k , the posterior remains Gaussian:

$$p(x_0 | x_t, k) = \mathcal{N}(x_0; m_k, S_k), \quad (21)$$

$$S_k = \left(\Sigma_k^{-1} + \frac{\alpha_t^2}{\sigma_t^2} I \right)^{-1}, \quad (22)$$

$$m_k = S_k \left(\Sigma_k^{-1} \mu_k + \frac{\alpha_t}{\sigma_t^2} x_t \right). \quad (23)$$

The responsibilities are

$$r_k(x_t) = \frac{\pi_k \mathcal{N}(x_t; \alpha_t \mu_k, \alpha_t^2 \Sigma_k + \sigma_t^2 I)}{\sum_j \pi_j \mathcal{N}(x_t; \alpha_t \mu_j, \alpha_t^2 \Sigma_j + \sigma_t^2 I)}. \quad (24)$$

Hence the full posterior is itself a Gaussian mixture:

$$p(x_0 | x_t) = \sum_{k=1}^K r_k(x_t) \mathcal{N}(x_0; m_k, S_k). \quad (25)$$

Posterior mean. The ideal denoiser in this case has a closed form:

$$m_{0|t}(x_t) := \mathbb{E}[x_0 | x_t] = \sum_{k=1}^K r_k(x_t) m_k. \quad (26)$$

For a fixed x_t , the posterior mean is a responsibility-weighted average of the component-wise posterior means, and can fall between mixture modes when the conditional posterior is multimodal.

Local MAP. Each component posterior has its mode at m_k . A local MAP predictor is obtained by selecting the component with the highest posterior peak density,

$$k^*(x_t) = \arg \max_k \frac{r_k(x_t)}{\sqrt{(2\pi)^d \det S_k}}, \quad x_0^*(t, x_t) = m_{k^*(x_t)}. \quad (27)$$

Unlike the posterior mean, this estimate is *mode-seeking* and stays in high-density regions.

DDIM iteration. Replacing the generic denoiser $m_{0|t}(x_t)$ in the DDIM update with either the posterior mean, local MAP yields 2 distinct variants of the reverse process:

$$x_{t-\Delta t} = g(m_{0|t}(x_t), x_t, \epsilon) \quad (\text{posterior mean}) \quad (28)$$

$$x_{t-\Delta t} = g(x_0^*(t, x_t), x_t, \epsilon) \quad (\text{local MAP}) \quad (29)$$

This toy setup makes explicit the distinction between *mean-based* denoising and *MAP-based* denoising.

Bias toward high-posterior modes. The above construction allows us to make precise in which sense local MAP is biased toward high-density regions of the posterior. For clarity, consider the special case where all mixture components share the same covariance, $\Sigma_k = \Sigma$, so that S_k and $\det S_k$ are independent of k . In this setting, the local MAP predictor simplifies to

$$k^*(x_t) = \arg \max_k r_k(x_t), \quad x_0^*(t, x_t) = m_{k^*(x_t)}, \quad (30)$$

that is, it selects the component with the largest responsibility. Equivalently, $x_0^*(t, x_t)$ is the maximizer of the joint posterior over the discrete–continuous pair (k, x_0) ,

$$(k^*, x_0^*) = \arg \max_{k, x_0} p(x_0, k | x_t) = \arg \max_k p(k | x_t), \quad (31)$$

where the maximizer over x_0 within each component is m_k . Thus local MAP coincides with the MAP estimator of the latent mixture index k (under 0–1 loss), followed by the corresponding component-wise posterior mode m_k .

Let $\mathcal{R}_k = \{x_t : k^*(x_t) = k\}$ denote the region of the diffusion state space where component k is selected. If we draw $x_t \sim p_t(x_t)$ and then apply local MAP, the probability that LMAPS outputs a sample associated with component k is

$$q_t(k) := \mathbb{P}[k^*(x_t) = k] = \int_{\mathcal{R}_k} p_t(x_t) dx_t. \quad (32)$$

By definition of $\mathcal{R}_k$, each $x_t \in \mathcal{R}_k$ satisfies $r_k(x_t) \geq r_j(x_t)$ for all $j \neq k$, so $\mathcal{R}_k$ collects those diffusion states where component k dominates the posterior responsibilities. Consequently, $q_t(k)$ is concentrated on modes with large posterior weight: whenever a component has small responsibilities $r_k(x_t)$ for almost all x_t , its region $\mathcal{R}_k$ has small measure and $q_t(k)$ is correspondingly small.

In the well-separated mixture regime, where the means $\{\mu_k\}$ are far apart relative to Σ and the diffusion noise, the posterior responsibilities $r_k(x_t)$ are nearly 0–1 valued. In this case, each region $\mathcal{R}_k$ is essentially the basin of attraction of mode k , and

$$q_t(k) \approx \int_{\text{basin}(k)} p_t(x_t) dx_t, \quad (33)$$

which is dominated by components with the highest posterior mass. Thus, even though LMAPS does not sample from the exact posterior mixture $\sum_k r_k(x_t) \mathcal{N}(m_k, S_k)$, its outputs are systematically biased toward high-posterior modes and avoid low-density regions between them. In contrast, posterior mean denoising yields mode-averaging estimates that may lie in low-density areas, and local posterior sampling explores all mixture components proportionally to their posterior mass. This toy example therefore formalizes the intuition that LMAPS interpolates between global MAP and posterior sampling by producing samples that concentrate on highly likely regions of the posterior while remaining stochastic along the diffusion trajectory.

B. Posterior covariance and asymptotic isotropy

Proposition B.1 (Gaussian prior). *Consider the forward noising process*

$$x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}), \quad (34)$$

and a Gaussian prior $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ with $\Sigma_0 \succ 0$. Then the posterior $p(x_0 | x_t)$ is Gaussian with covariance

$$\Sigma_{0|t} = (\Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I})^{-1} \preceq \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I}. \quad (35)$$

Moreover, as $\sigma_t^2/\alpha_t^2 \rightarrow 0$, the covariance admits the asymptotic expansion

$$\Sigma_{0|t} = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I} + \mathcal{O}\left(\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2\right), \quad (36)$$

so the leading term is isotropic and any anisotropy appears only in higher-order corrections.

Proof. Since (x_0, x_t) is jointly Gaussian under the model

$$x_0 \sim \mathcal{N}(\mu_0, \Sigma_0), \quad x_t | x_0 \sim \mathcal{N}(\alpha_t x_0, \sigma_t^2 \mathbb{I}), \quad (37)$$

the posterior $p(x_0 | x_t)$ is Gaussian. Equivalently, we can view x_t as a linear observation of x_0 with observation matrix $H = \alpha_t \mathbb{I}$ and noise covariance $R = \sigma_t^2 \mathbb{I}$. The standard linear Gaussian posterior formula (or Kalman update) yields

$$\Sigma_{0|t}^{-1} = \Sigma_0^{-1} + H^\top R^{-1} H = \Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}, \quad (38)$$

so

$$\Sigma_{0|t} = \left(\Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} \right)^{-1}. \quad (39)$$

For the Loewner-order upper bound, note that $\Sigma_0^{-1} \succeq 0$, so

$$\Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} \succeq \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}. \quad (40)$$

For positive definite matrices, the matrix inverse is order-reversing: if $A \succeq B \succ 0$, then $A^{-1} \preceq B^{-1}$. Applying this with $A = \Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}$ and $B = \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}$ gives

$$\Sigma_{0|t} = A^{-1} \preceq B^{-1} = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I}. \quad (41)$$

For the asymptotic expansion, factor out the isotropic term:

$$\Sigma_{0|t} = \left(\Sigma_0^{-1} + \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} \right)^{-1} = \frac{\sigma_t^2}{\alpha_t^2} \left(\mathbb{I} + \frac{\sigma_t^2}{\alpha_t^2} \Sigma_0^{-1} \right)^{-1}. \quad (42)$$

Let $\varepsilon_t := \frac{\sigma_t^2}{\alpha_t^2}$. For $\varepsilon_t \rightarrow 0$ we may use the Neumann series

$$\left(\mathbb{I} + \varepsilon_t \Sigma_0^{-1} \right)^{-1} = \mathbb{I} - \varepsilon_t \Sigma_0^{-1} + \mathcal{O}(\varepsilon_t^2), \quad (43)$$

which yields

$$\Sigma_{0|t} = \varepsilon_t \left(\mathbb{I} - \varepsilon_t \Sigma_0^{-1} + \mathcal{O}(\varepsilon_t^2) \right) = \varepsilon_t \mathbb{I} + \mathcal{O}(\varepsilon_t^2) = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I} + \mathcal{O}\left(\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2\right). \quad (44)$$

The leading term is therefore isotropic, and any anisotropy is of order $\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2$. $\square$

Proposition B.2 (General prior and asymptotic isotropy). *Assume the forward noising process*

$$x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}), \quad (45)$$

and an arbitrary prior density $p(x_0)$ such that $-\log p(x_0)$ is twice continuously differentiable. Then the negative log-posterior is

$$-\log p(x_0 | x_t) = -\log p(x_0) + \frac{1}{2\sigma_t^2} \|x_t - \alpha_t x_0\|^2 + \text{const}, \quad (46)$$

and its Hessian with respect to x_0 satisfies

$$\nabla_{x_0}^2 [-\log p(x_0 | x_t)] = \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} + H_{\text{prior}}(x_0), \quad (47)$$

where $H_{\text{prior}}(x_0) := \nabla_{x_0}^2 [-\log p(x_0)]$. If $H_{\text{prior}}(x_0)$ is bounded in operator norm on the region of interest, then as $\sigma_t^2 \rightarrow 0$ (and $\alpha_t \rightarrow 1$), the local Gaussian (Laplace) approximation to $p(x_0 | x_t)$ has covariance

$$\Sigma_{0|t}^{\text{Laplace}}(x_0) = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I} + \mathcal{O}\left(\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2\right), \quad (48)$$

and is therefore asymptotically isotropic as $t \rightarrow 0$.

Proof. The expression for $-\log p(x_0 | x_t)$ follows directly from Bayes' rule and the Gaussian likelihood:

$$p(x_t | x_0) \propto \exp\left(-\frac{1}{2\sigma_t^2} \|x_t - \alpha_t x_0\|^2\right). \quad (49)$$

Taking the Hessian with respect to x_0 gives

$$\nabla_{x_0}^2 \left[\frac{1}{2\sigma_t^2} \|x_t - \alpha_t x_0\|^2 \right] = \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}, \quad (50)$$

while the prior contributes

$$\nabla_{x_0}^2 [-\log p(x_0)] = H_{\text{prior}}(x_0). \quad (51)$$

Therefore,

$$H_{\text{post}}(x_0) := \nabla_{x_0}^2 [-\log p(x_0 | x_t)] = \frac{\alpha_t^2}{\sigma_t^2} \mathbb{I} + H_{\text{prior}}(x_0). \quad (52)$$

Assume $\|H_{\text{prior}}(x_0)\|_{\text{op}} \leq C$ for some constant C . Then, in the regime $\sigma_t^2 \rightarrow 0$ and $\alpha_t \rightarrow 1$, the dominant term in $H_{\text{post}}(x_0)$ is the isotropic matrix $\frac{\alpha_t^2}{\sigma_t^2} \mathbb{I}$. Define again $\varepsilon_t := \frac{\sigma_t^2}{\alpha_t^2}$ and write

$$H_{\text{post}}(x_0) = \frac{\alpha_t^2}{\sigma_t^2} \left(\mathbb{I} + \varepsilon_t H_{\text{prior}}(x_0) \right). \quad (53)$$

The local Gaussian (Laplace) approximation uses $\Sigma_{0|t}^{\text{Laplace}}(x_0) = H_{\text{post}}(x_0)^{-1}$. Applying the Neumann series to $(\mathbb{I} + \varepsilon_t H_{\text{prior}}(x_0))^{-1}$ for small ε_t yields

$$(\mathbb{I} + \varepsilon_t H_{\text{prior}}(x_0))^{-1} = \mathbb{I} - \varepsilon_t H_{\text{prior}}(x_0) + \mathcal{O}(\varepsilon_t^2), \quad (54)$$

so

$$\Sigma_{0|t}^{\text{Laplace}}(x_0) = \varepsilon_t \left(\mathbb{I} - \varepsilon_t H_{\text{prior}}(x_0) + \mathcal{O}(\varepsilon_t^2) \right) = \varepsilon_t \mathbb{I} + \mathcal{O}(\varepsilon_t^2) = \frac{\sigma_t^2}{\alpha_t^2} \mathbb{I} + \mathcal{O}\left(\left(\frac{\sigma_t^2}{\alpha_t^2}\right)^2\right). \quad (55)$$

Thus the leading term of the local covariance is isotropic, and any anisotropy is of strictly higher order in σ_t^2/α_t^2 . $\square$

C. Tempered local posterior interpretation

The tunable parameter k in Equation (14) can be interpreted as changing the temperature of the local posterior. Under the isotropic approximation,

$$p_k(x_0 | x_t) = \mathcal{N}\left(x_0; m_{0|t}, \frac{k}{\text{SNR}} \mathbb{I}\right) \propto \exp\left(-\frac{\text{SNR}}{2k} \|x_0 - m_{0|t}\|^2\right). \quad (56)$$

If $p_1(x_0 | x_t)$ denotes the same Gaussian approximation with $k = 1$, then

$$p_k(x_0 | x_t) \propto p_1(x_0 | x_t)^{1/k}. \quad (57)$$

Therefore the corresponding local posterior satisfies

$$p_k(x_0 | x_t, y) \propto p(y | x_0) p_1(x_0 | x_t)^{1/k}. \quad (58)$$

Thus $k > 1$ flattens the local diffusion prior and gives relatively more weight to measurement consistency, while $k < 1$ sharpens the prior and makes the update more conservative around $m_{0|t}$. In this sense, k implements a tempered local posterior rather than changing the forward measurement model.

D. Sampling Efficiency

We present a comparison of sampling times among LMAPS, DAPS, and SITCOM. Among them, SITCOM and DAPS achieve the third- and second-best results, respectively, while LMAPS demonstrates the best performance with lower computation time.

Table 3. Sampling time of LMAPS on Deblurring tasks with FFHQ 256. The non-parallel single-image sampling time on the FFHQ 256 dataset using one NVIDIA A6000 GPU. NFE refers to diffusion timesteps, while optimization steps refer to inner loop optimizations in respective methods.

Configuration	ODE Steps	Optimization Steps	NFE	Second/Image	LPIPS
DAPS	5	100	200	110	0.165
SITCOM	–	30	600	73	0.172
DPS	–	–	1000	138	0.219
MAPS	–	100	200	61	0.158
	–	10	100	6	0.190
	–	100	20	6	0.156
	–	20	100	6	0.176
	–	20	20	2	0.180

E. Experiment details

E.1. Dataset details

For scientific inverse problems, we adopt fluorescence microscopy images from InverseBench (Zheng et al., 2025) on linear inverse scattering tasks, General Relativistic MagnetoHydroDynamic (GRMHD) (Mizuno, 2022) on black hole imaging, and multi-coil raw k -space data from the fastMRI knee dataset (Zbontar et al., 2018) on CS-MRI.

E.2. Inverse problem details

Baselines from DAPS (Zhang et al., 2025a). For image restoration tasks include: (1) super-resolution, (2) Gaussian deblurring, (3) motion deblurring, (4) inpainting (with a box mask), and (5) inpainting (with a 70% random mask), (6) phase retrieval, (7) high dynamic range (HDR) reconstruction, (8) nonlinear deblurring, we follow the same experimental setup as in DAPS.

InverseBench (Zheng et al., 2025). For scientific inverse problems, we adopt the setting introduced in InverseBench.

JPEG Restoration. We address JPEG restoration with quality factors of $QF = 5$.

Quantization. We model quantization by discretizing the measurement into $2^{n_{\text{bits}}}$ uniformly spaced levels. Formally, the forward operator is defined as

$$\mathcal{H}(x) = \frac{\lfloor x \cdot (2^{n_{\text{bits}}} - 1) + 0.5 \rfloor}{2^{n_{\text{bits}}} - 1}, \quad (59)$$

which rounds the input x to the nearest quantization level. In this work, we focus on the challenging case of 2-bit quantization, where only four distinct measurement levels are available, significantly reducing precision and making accurate reconstruction more difficult.

E.3. Baseline details

For SITCOM (Alkhouri et al., 2024), we use the hyperparameter configuration recommended in the original paper, with $N = 20$ and $K = 30$, resulting in 600 NFEs and requiring gradient computation with respect to the U-Net.

For DMPlug (Wang et al., 2024), we set epoch = 1000 for SR, Inpainting (Random) and Nonlinear Deblurring, other parameters are the same as suggested in the original paper.

For MMPS (Rozet et al., 2024), we set steps as 100, the maximum number of iterations $N = 5$.

For DMAP (Xu et al., 2025), we report the available baseline results on linear image restoration tasks under the same FFHQ/ImageNet evaluation protocol.

For non-differentiable inverse problems, we use IIGDM (Song et al., 2023b) as our baseline approaches, we adopt $NFE = 100$ as suggested in the original paper.

Other baselines we adopt the same reported results as in DAPS (Zhang et al., 2025a) and InverseBench (Zheng et al., 2025).

Tasks	Annealing step	Gradient step	Learning rate η	k_1	k_2
SR 4×	200	100	0.05	0.15	20
Inpaint (Box)	200	100	0.02	0.5	50
Inpaint (Random)	200	100	0.01	0.22	100
Gaussian Deblurring	200	100	0.01	0.22	100
Motion Deblurring	200	100	0.01	0.25	100
Phase Retrieval	200	100	0.1	10	0.3
Nonlinear Deblurring	200	100	0.02	0.05	1
High Dynamic Range	200	100	0.04	0.2	10
JPEG Restoration	200	100	0.2	0.5	5
Quantization	200	20	0.2	0.5	5
LIS (NR=360)	200	50	1	0	5000
LIS (NR=180)	200	50	1	0	10000
LIS (NR=60)	200	50	1	0	30000
CS-MRI (4×, noiseless)	200	100	0.01	0	100
CS-MRI (4×, raw)	200	100	0.01	0.4	150
CS-MRI (8×, noiseless)	200	100	0.01	0.4	150
CS-MRI (8×, raw)	200	100	0.01	0.4	150
Black Hole (ratio=100%)	100	200	0.01	0.1	0.01
Black Hole (ratio=10%)	100	200	0.005	0.1	0.03
Black Hole (ratio=3%)	100	200	0.01	0.05	0.05

Table 4. Complete List of hyper-parameters of LMAPS for different inverse problems.

E.4. Complete List of hyper-parameters

We provide complete list of hyper-parameters of LMAPS for different inverse problems in Table 4.

F. Additional experiment results

F.1. Scientific inverse problems

We present additional evaluation metrics on linear inverse scattering in Table 5, compressed sensing MRI in Table 6, and black hole imaging in Table 7.

F.2. Additional visualization

Additional visualization are presented in Figs. 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

F.3. Comparison between analytical solution and gradient descent for solving LMAPS

We present the comparison between analytical solution and gradient descent for solving LMAPS in Table 8. The results demonstrate that the analytical solution closely matches the gradient-descent-based optimizer, with only minor differences in reconstruction metrics. This confirms that our analytical formulation is a reliable and efficient approximation for solving the LMAPS objective.

F.4. Additional results on Nonlinear Deblurring

For Nonlinear Deblurring, the forward operator call is relatively expensive. The results on solving Nonlinear Deblurring with different annealing step and gradient step are shown in Table 9. LMAPS can achieve competitive performance with only 100 annealing steps and 20 gradient steps.

Table 5. Results on Linear inverse scattering. PSNR and SSIM of different algorithms on linear inverse scattering. Noise level $\sigma_y = 10^{-4}$.

Number of receivers	360		180		60	
Methods	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Traditional						
FISTA-TV	32.126 (2.139)	0.979 (0.009)	26.523 (2.678)	0.914 (0.040)	20.938 (2.513)	0.709 (0.103)
PnP diffusion prior						
DDRM	32.598 (1.825)	0.929 (0.012)	28.080 (1.516)	0.890 (0.019)	20.436 (1.210)	0.545 (0.037)
DDNM	36.381 (1.098)	0.935 (0.017)	35.024 (0.993)	0.895 (0.027)	29.235 (3.376)	0.917 (0.022)
IIGDM	27.925 (3.211)	0.889 (0.072)	26.412 (3.430)	0.816 (0.114)	20.074 (2.608)	0.540 (0.198)
DPS	32.061 (2.163)	0.846 (0.127)	31.798 (2.163)	0.862 (0.123)	27.372 (3.415)	0.813 (0.133)
LGD	27.901 (2.346)	0.812 (0.037)	27.837 (3.031)	0.803 (0.034)	20.491 (3.031)	0.552 (0.077)
DiffPIR	34.241 (2.310)	0.988 (0.006)	34.010 (2.269)	0.987 (0.006)	26.321 (3.272)	0.918 (0.028)
PnP-DM	33.914 (2.054)	0.988 (0.006)	31.817 (2.073)	0.981 (0.008)	24.715 (2.874)	0.909 (0.046)
DAPS	34.641 (1.693)	0.957 (0.006)	33.160 (1.704)	0.944 (0.009)	25.875 (3.110)	0.885 (0.030)
RED-diff	36.556 (2.292)	0.981 (0.005)	35.411 (2.166)	0.984 (0.004)	27.072 (3.330)	0.935 (0.037)
FPS	33.242 (1.602)	0.870 (0.026)	29.624 (1.651)	0.710 (0.040)	21.323 (1.445)	0.460 (0.030)
MCG-diff	30.937 (1.964)	0.751 (0.029)	28.057 (1.672)	0.631 (0.042)	21.004 (1.571)	0.445 (0.028)
LMAPS	38.074 (1.905)	0.994 (0.001)	37.188 (1.815)	0.990 (0.001)	30.759 (3.539)	0.967 (0.211)

Table 6. Results on compressed sensing MRI. Mean and standard deviation are reported over 94 test cases. Underline: the best across all methods. Bold: the best across PnP DM methods.

Methods	$\times 4$ Simulated (noiseless)			$\times 4$ Raw			$\times 8$ Simulated (noiseless)			$\times 8$ Raw		
	PSNR $\uparrow$	SSIM $\uparrow$	Data misfit $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Data misfit $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Data misfit $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Data misfit $\downarrow$
Traditional												
Wavelet+ ℓ_1	29.45 (1.776)	0.690 (0.121)	0.306 (0.049)	26.47 (1.508)	0.598 (0.122)	31.601 (15.286)	25.97 (1.761)	0.575 (0.105)	0.318 (0.042)	24.08 (1.602)	0.511 (0.106)	22.362 (10.733)
TV	27.03 (1.635)	0.518 (0.123)	5.748 (1.283)	26.22 (1.578)	0.509 (0.123)	32.269 (15.414)	24.12 (1.900)	0.432 (1.112)	5.087 (1.049)	23.70 (1.857)	0.427 (0.112)	23.048 (10.854)
End-to-end												
Residual UNet	32.27 (1.810)	0.808 (0.080)	–	31.70 (1.970)	0.785 (0.095)	–	29.75 (1.675)	0.750 (0.088)	–	29.36 (1.746)	0.733 (0.100)	–
E2E-VarNet	33.40 (2.097)	0.836 (0.079)	–	31.71 (2.540)	0.756 (0.102)	–	30.67 (1.761)	0.769 (0.085)	–	30.45 (1.940)	0.736 (0.103)	–
PnP diffusion prior												
CSGM	28.78 (6.173)	0.710 (0.147)	1.518 (0.433)	25.17 (6.246)	0.582 (0.167)	31.642 (15.382)	26.15 (6.383)	0.625 (0.158)	1.142 (1.078)	21.17 (8.314)	0.425 (0.192)	22.088 (10.740)
ScoreMRI	25.97 (1.681)	0.468 (0.087)	10.828 (1.731)	25.60 (1.618)	0.463 (0.086)	33.697 (15.209)	25.20 (1.526)	0.405 (0.079)	8.360 (1.381)	24.74 (1.481)	0.403 (0.080)	24.028 (10.663)
RED-diff	29.36 (7.710)	0.733 (0.131)	0.509 (0.077)	28.71 (2.755)	0.626 (0.126)	31.591 (15.368)	26.76 (6.969)	0.647 (0.124)	0.485 (0.068)	27.33 (2.441)	0.563 (0.117)	22.336 (10.838)
DiffPIR	28.31 (1.598)	0.632 (0.107)	10.545 (2.466)	27.60 (1.470)	0.624 (0.111)	34.015 (15.522)	26.78 (1.556)	0.588 (0.113)	7.787 (1.741)	26.26 (1.458)	0.590 (0.113)	24.208 (10.922)
DPS	26.13 (4.247)	0.620 (0.105)	9.092 (2.925)	25.83 (2.197)	0.548 (0.116)	35.009 (15.967)	22.82 (4.777)	0.536 (0.111)	6.737 (1.928)	23.00 (3.205)	0.507 (0.109)	24.842 (11.263)
DAPS	31.48 (1.988)	0.762 (0.089)	1.566 (0.390)	28.61 (2.197)	0.689 (0.102)	31.115 (15.497)	29.01 (1.712)	0.681 (0.098)	1.280 (0.301)	27.10 (2.034)	0.629 (0.107)	22.729 (10.926)
PnP-DM	31.80 (3.473)	0.780 (0.096)	4.701 (0.675)	27.62 (3.425)	0.679 (0.117)	32.261 (15.169)	29.33 (3.081)	0.704 (0.105)	3.421 (0.504)	25.28 (3.102)	0.607 (0.117)	22.879 (10.712)
LMAPS	32.83 (2.581)	0.740 (0.117)	3.500 (0.544)	28.77 (1.813)	0.656 (0.102)	32.476 (15.303)	30.50 (2.181)	0.660 (0.116)	2.565 (0.399)	27.43 (1.689)	0.600 (0.109)	23.021 (10.804)

F.5. Ablation on measurement-conditioning strength

In Equation (15), k_2 controls the strength of the measurement-conditioned term and thus the extent of mode-seeking in $p(x_0 | x_t, y)$. To isolate this effect, we ablate k_2 on FFHQ super-resolution with 20 NFEs while solving

$$x_0^* = \arg \min_{x_0} \left(1 - \frac{\sigma_t^2}{\sigma_t^2 + k_1^2} \right) \frac{1}{2} \|x_0 - m_{0|t}\|^2 + \frac{\sigma_t^2}{\sigma_t^2 + k_1^2} k_2 \|y - \mathcal{H}(x_0)\|^2. \quad (60)$$

Table 10 shows that increasing k_2 consistently improves all reconstruction metrics. The $k_2 = 0$ setting removes the measurement term and performs poorly, confirming that explicitly optimizing the measurement-conditioned local MAP objective is essential.

G. Licenses

FFHQ Dataset. We use the Flickr-Faces-HQ (FFHQ) dataset released by NVIDIA under the Creative Commons BY-NC-SA 4.0 license. The dataset is intended for non-commercial research purposes only. More details are available at: <https://github.com/NVlabs/ffhq-dataset>.

ImageNet Dataset. The ImageNet dataset is used under the terms of its academic research license. Access requires agreement to ImageNet’s data use policy, and redistribution is not permitted. More information is available at: <https://image-net.org/download>.

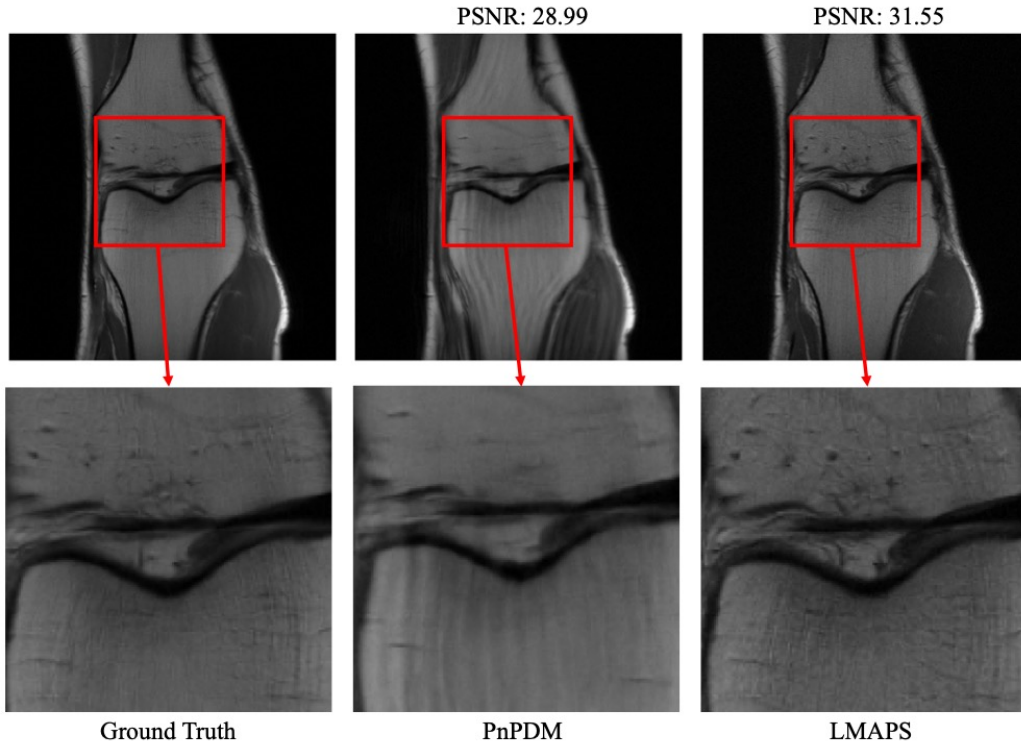


Figure 5. Visualization of CS-MRI restoration ($4\times$ raw).

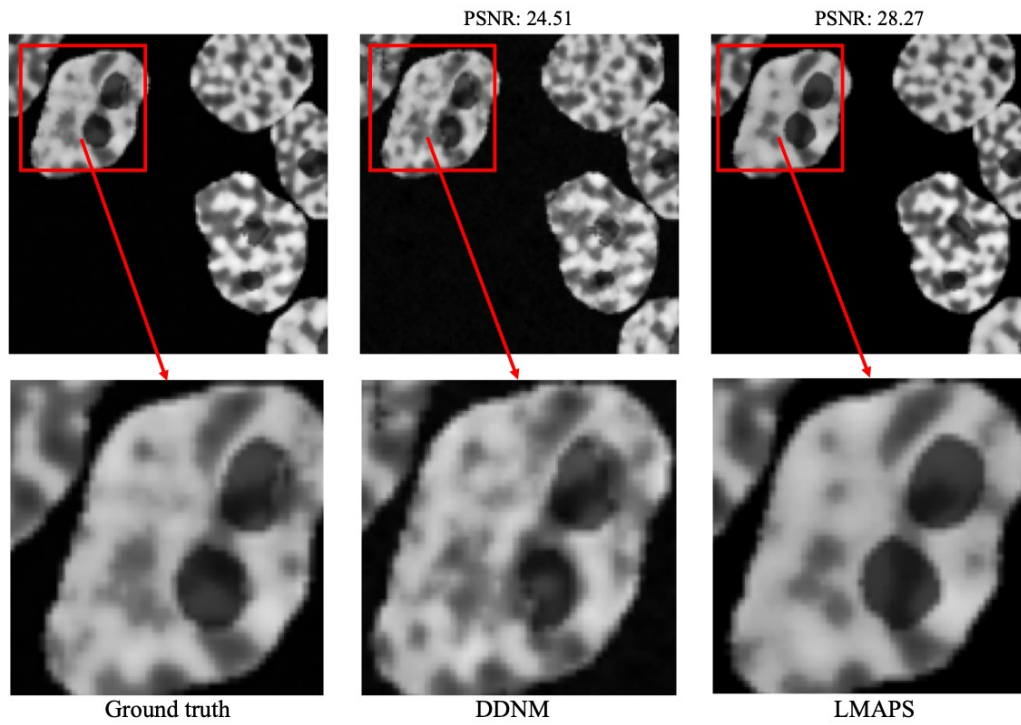


Figure 6. Visualization of Linear Inverse Scattering (Number of receivers = 60).



Figure 7. Visualization for solving JPEG restoration ($QF=5$, $\sigma_y = 0.05$). Top: degraded images; bottom: generated images.



Figure 8. Visualization for solving Quantization (2 bit). Top: degraded images; bottom: generated images.

Table 7. Results on black hole imaging. PSNR and Chi-squared of different algorithms on black hole imaging. Gain and phase noise and thermal noise are added based on EHT library.

Methods	3%				10%				100%			
	PSNR	Blur PSNR	χ^2_{exp}	χ^2_{logca}	PSNR	Blur PSNR	χ^2_{exp}	χ^2_{logca}	PSNR	Blur PSNR	χ^2_{exp}	χ^2_{logca}
Traditional												
SMILI	18.51 (1.39)	23.08 (2.12)	1.478 (0.428)	4.348 (3.827)	20.85 (2.90)	25.24 (3.86)	1.209 (0.169)	21.788 (12.491)	22.67 (3.13)	27.79 (4.02)	1.878 (0.952)	17.612 (10.299)
EHT-Imaging	21.72 (3.39)	25.66 (5.04)	1.507 (0.485)	1.695 (0.539)	22.67 (3.46)	26.66 (3.93)	1.166 (0.156)	1.240 (0.205)	24.28 (3.63)	28.57 (4.52)	1.251 (0.250)	1.259 (0.316)
PnP diffusion prior												
DPS	24.20 (3.72)	30.83 (5.58)	8.024 (24.336)	5.007 (5.750)	24.36 (3.72)	30.79 (5.75)	13.052 (43.087)	6.614 (26.789)	25.86 (3.90)	32.94 (6.19)	8.759 (37.784)	5.456 (24.185)
LGD	22.51 (3.76)	28.50 (5.49)	15.825 (16.838)	12.862 (12.663)	22.08 (3.75)	27.48 (5.09)	10.775 (21.684)	13.375 (56.397)	21.22 (3.64)	26.06 (4.98)	13.239 (17.231)	13.233 (39.107)
RED-diff	20.74 (2.62)	26.10 (3.35)	6.713 (6.925)	9.128 (19.052)	22.53 (3.02)	27.67 (4.53)	2.488 (2.925)	4.916 (13.221)	23.77 (4.13)	29.13 (6.22)	1.853 (0.938)	2.050 (2.361)
PnPDM	24.25 (3.45)	30.49 (4.93)	2.201 (1.352)	1.668 (0.551)	24.57 (3.47)	30.80 (5.22)	1.433 (0.417)	1.336 (0.478)	26.07 (3.70)	32.88 (6.02)	1.311 (0.195)	1.199 (0.221)
DAPS	23.54 (3.28)	29.48 (4.88)	3.647 (3.287)	2.329 (1.354)	23.99 (3.56)	30.16 (5.13)	1.545 (0.705)	2.253 (9.903)	25.60 (3.64)	32.78 (5.68)	1.300 (0.324)	1.229 (0.532)
DiffPIR	24.12 (3.25)	30.45 (4.88)	14.085 (14.105)	10.545 (8.860)	23.84 (3.59)	30.04 (5.03)	5.374 (3.733)	5.205 (5.556)	25.01 (4.64)	31.86 (6.56)	3.271 (1.623)	2.970 (1.202)
LMAPS	24.66 (4.02)	29.94 (5.17)	1.497 (0.394)	4.695 (1.420)	24.84 (3.695)	29.98 (5.144)	1.671 (0.521)	4.460 (1.555)	26.79 (3.78)	32.95 (5.41)	1.512 (0.474)	4.622 (1.455)

Table 8. Comparison between analytical solution and gradient descent for solving LMAPS, LMAPS-GD represents solving LMAPS with gradient descent, LMAPS-A refers to solving LMAPS with analytical solution.

Task	Method	FFHQ			ImageNet		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
SR $4\times$	LMAPS-GD	30.74	0.869	0.165	26.72	0.739	0.242
	LMAPS-A	30.31	0.860	0.161	26.39	0.723	0.252
Inpaint (Box)	LMAPS-GD	25.02	0.876	0.108	21.25	0.803	0.204
	LMAPS-A	25.35	0.871	0.120	21.15	0.796	0.216

Table 9. Solving Nonlinear Deblurring with different annealing step and gradient step.

Annealing steps	Gradient steps	FFHQ			ImageNet		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
200	200	29.93 $\pm$ 1.83	0.855$\pm$0.035	0.150$\pm$0.034	28.03 $\pm$ 3.62	0.774 $\pm$ 0.099	0.183 $\pm$ 0.065
100	20	27.58 $\pm$ 1.878	0.814 $\pm$ 0.024	0.200 $\pm$ 0.040	26.15 $\pm$ 3.24	0.729 $\pm$ 0.118	0.257 $\pm$ 0.079

 Table 10. Ablation of k_2 on FFHQ super-resolution with 20 NFEs.

k_2	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
0	10.514	0.391	0.613	156.00
0.001	16.285	0.517	0.498	145.72
0.005	18.879	0.584	0.424	136.15
0.01	20.102	0.615	0.392	130.89
0.05	22.999	0.688	0.319	116.82
0.1	24.198	0.717	0.291	110.15
0.5	26.755	0.777	0.234	95.22
1	27.695	0.799	0.211	87.78

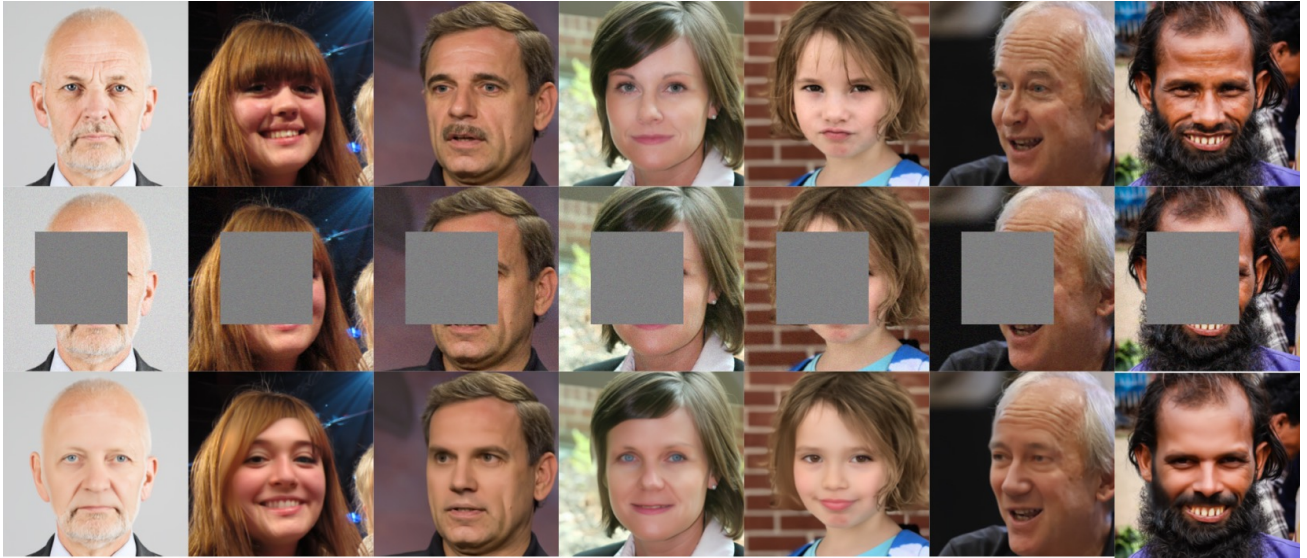


Figure 9. Visualization for solving Inpaint (Box). Top: ground truth; middle: degraded images; bottom: generated images.



Figure 10. Visualization for solving HDR. Top: ground truth; middle: degraded images; bottom: generated images.



Figure 11. Visualization for solving Deblurring. Top: ground truth; middle: degraded images; bottom: generated images.



Figure 12. Visualization for solving Super-Resolution. Top: ground truth; middle: degraded images; bottom: generated images.



Figure 13. Visualization for solving Nonlinear Deblurring. Top: ground truth; middle: degraded images; bottom: generated images.

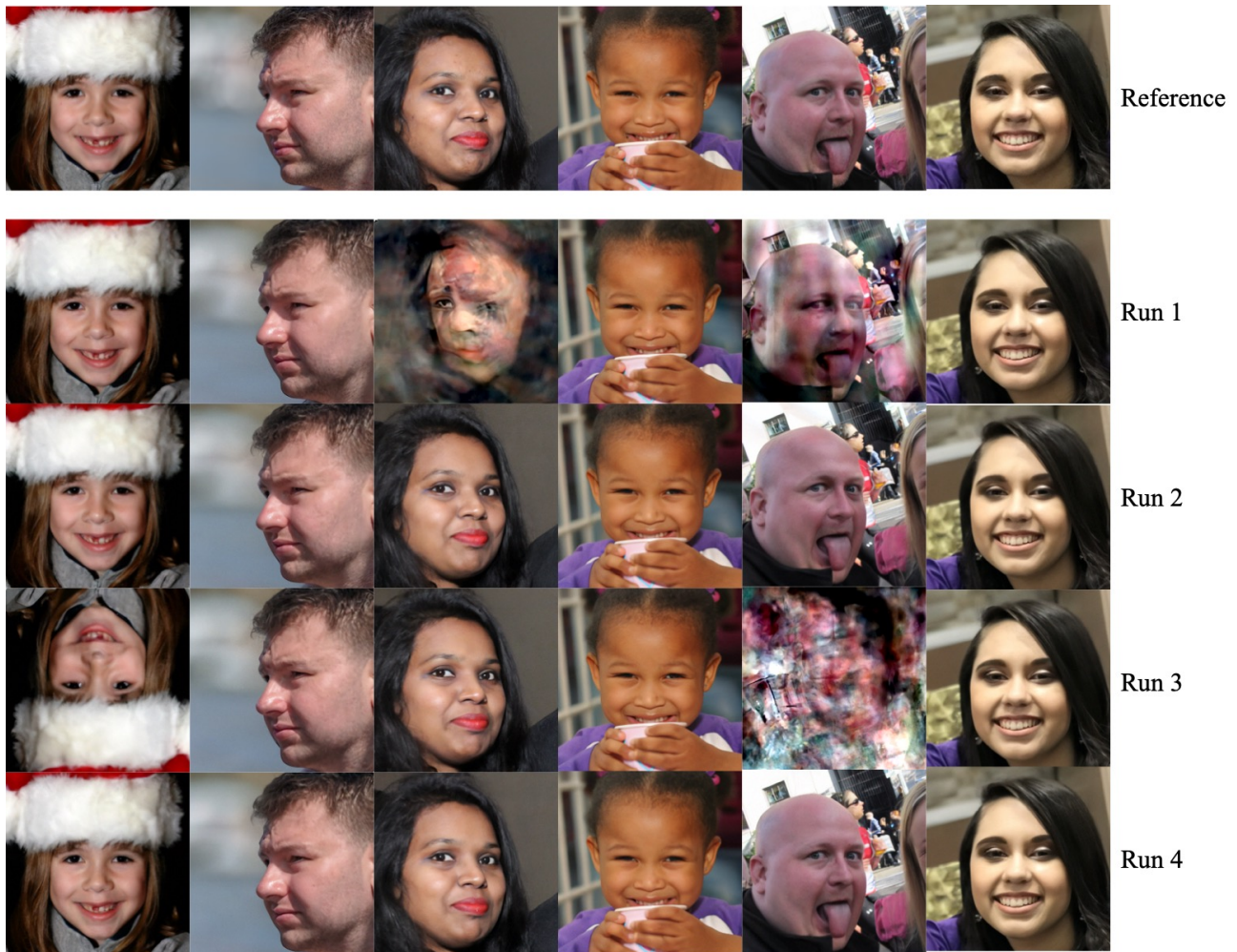


Figure 14. Visualization for solving Phase retrieval.